

Assignment 2

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0.1 Instructions

This problem set is based on Lecture 7. It contains five hard, multi-step questions on interest-rate models, no-arbitrage short-rate models, Hull-White calibration, future discount curves, and callable bond valuation. The questions emphasize conceptual understanding and professional fixed-income reasoning.

For each question:

1. Read the scenario carefully.
2. Identify whether today's discount curve is sufficient or whether future interest-rate dynamics matter.
3. Explain the valuation logic before doing any calculations.
4. Show formulas only where they clarify the economic reasoning.

1 Lecture 7: Interest-Rate Models

1.1 Question 1

A junior analyst is valuing two securities for a client:

Security	Description	Cash-flow feature
A	6-year Treasury note with a 4.25 percent annual coupon	Fixed cash flows
B	6-year agency bond with a 4.25 percent annual coupon, callable at par after year 2	Issuer may redeem early

The analyst proposes using a Hull-White short-rate model for both securities because “interest rates are uncertain.”

Answer the following:

- Write the basic discounted-cash-flow expression for valuing Security A using discount factors.
- Explain why a bootstrapped discount curve is usually enough for Security A.
- Explain why calibrating Hull-White to today’s curve should reproduce the same time-0 value for Security A.
- Explain why Security B may require a short-rate model even though its coupon rate is the same as Security A.
- Identify the main mistake in the analyst’s statement.

1.2 Question 2

A rates desk observes the following market discount factors at time 0:

Maturity T	$P(0, T)$
1	0.9600
2	0.9200
3	0.8750
4	0.8300

Maturity T	$P(0, T)$
5	0.7850

The desk calibrates a one-factor Hull-White model to this initial curve. Later, in a simulated future state at time $t = 2$, the model generates the following future discount factors:

Future maturity T	$P(2, T)$
3	0.9650
4	0.9250
5	0.8800

Answer the following:

- Explain the difference between $P(0, T)$ and $P(2, T)$.
- Which set of discount factors is observed or calibrated at time 0, and which set is model-generated?
- Use the time-0 discount factors to value a 5-year zero-coupon bond paying USD 1,000 at maturity.
- Use the future discount factors to value, at time 2, remaining cash flows of USD 50 at years 3 and 4 and USD 1,050 at year 5.
- Explain why the future value in part d matters for callable bond valuation.

1.3 Question 3

A portfolio manager is reviewing a one-factor Hull-White model:

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t.$$

The manager says: “The model has only two real parameters, a and σ , so it cannot possibly fit an entire yield curve.”

Answer the following:

- Explain what a controls.

- b. Explain what σ controls.
- c. Explain why $\theta(t)$ is not just one scalar parameter.
- d. Explain how $\theta(t)$ is used in calibration.
- e. Explain why a and σ are often calibrated to option prices rather than plain bond prices.

1.4 Question 4

A callable bond has three annual remaining cash flows after the first call date:

Payment date	Cash flow
3	USD 60
4	USD 60
5	USD 1,060

At the call date $t = 2$, the call price is USD 1,020. In one simulated interest-rate state, the model-generated future discount factors are:

Future maturity T	$P(2, T)$
3	0.9850
4	0.9550
5	0.9200

In another simulated state, the model-generated future discount factors are:

Future maturity T	$P(2, T)$
3	0.9300
4	0.8650
5	0.8000

Answer the following:

- a. Compute the continuation value at time 2 in the first state.
- b. Decide whether the issuer calls the bond in the first state.

- c. Compute the bondholder's value at time 2 in the first state.
- d. Repeat parts a-c for the second state.
- e. Explain the economic intuition behind the different call decisions.

1.5 Question 5

An investment committee is comparing four instruments:

Instrument	Key feature
1	Fixed-rate Treasury bond
2	Callable corporate bond
3	Interest-rate cap
4	Mortgage-backed security

The committee wants to know when a short-rate model is necessary and when it is unnecessary.

Answer the following:

- a. Classify each instrument as usually requiring only today's discount curve or requiring future rate dynamics.
- b. For each instrument that requires future rate dynamics, explain what future-rate-dependent feature drives the need for a model.
- c. Explain why a yield curve is a snapshot but a short-rate model is a scenario generator.
- d. Explain why mortgage-backed securities require more than just interest-rate modeling.
- e. State one practical limitation of using a one-factor short-rate model for risk management.